

M01 / STUDY COMPANION

M01 Quick Revision Book

Standalone Zero-to-Job-Ready Data Engineer 2026

Study, practise, check and explain.

A complete learning route

Start with the lesson video. Read the worked explanation beside it, pause to reproduce the example, then solve the supplied task. The books contain your study notes; keep your own work and results as evidence.

Use the protected answers after saving your first attempt. A changed retry shows what you can now do independently. There is no forced timer during learning.

One course. M00-M19. 264 lessons and 50 project phases.

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Read this first

QUICK REVISION BOOK

Use this only after you have studied the learner book. Each lesson keeps the canonical language and core proof, plus one recall task. This is not a replacement for hands-on practice.

Coverage: 15 canonical lessons

Route: learner book -> practice -> revision -> closed-book test -> protected answers -> changed retry.

How to revise

Read the lesson card, close the page, explain it in your own words, then answer the recall task. If recall is weak, return to that lesson only - not the entire module.

DE01-01 | What data is: tables, records, fields and data types

60-second recall

Remember: Meaning Table: A grid of named fields and records Record / row: One entry; always state what it represents Field / column: One kind of information collected for each entry Cell and value: A cell is a position; its value is the information there Header: Names fields; it is not another data record Type: The intended category of a value, not only its appearance Identifier: A label such as 0007 or A017; text can preserve its identity Missing: No supplied answer; not automatically zero or FALSE Reconciliation: Compare results that should agree; remember the check's limits

What understanding looks like

You can read a new row aloud, identify fields and records, explain types, calculate an amount, preserve an unknown status and check the result. After a mistake, you can explain the correction and solve the changed requirement without copying the first answer.

[Refresh source: What is Data? An Introduction to Data Terminology and Concepts](#)

Recall task

Changed task: answer a different question

The owner now asks: "What is the value of orders whose payment status is known, whether paid or unpaid?" Include TRUE and FALSE. Exclude missing. This differs from asking only for paid orders.

order_id: item | quantity | unit_price | ordered_on | is_paid A015: Pencil | 5 | 8.00 | 2026-09-05 | TRUE A016: Eraser | 2 | 5.00 | 2026-09-05 | FALSE A017: Bag | 1 | 300.00 | 2026-09-05 | (missing)

State the row meaning and counts. Calculate the row amounts.

DE01-02 | Grain: what one row represents

60-second recall

State the grain before counting, grouping or combining data. Rows, orders, lines and items are different counts. A repeated identifier can be legitimate at a finer grain. Summarise each value according to the thing it describes.

[Refresh source: What is Data? An Introduction to Data Terminology and Concepts](#)

Recall task

Order A101 has three lines: 1 bag at 200.00, 2 pencils at 8.00 and 1 ruler at 20.00. There is one 10.00 delivery fee. Calculate the goods and payable, then explain why a result with three rows would not yet be the requested one-row-per-order summary.

DE01-03 | Keys, duplicates and missing values

60-second recall

Define a key at the grain you intend. Check missing key parts and repeated keys. Distinguish identical rows from conflicting rows. Preserve unknown values and report uncertainty instead of inventing a convenient answer.

[Refresh source: Data Accuracy and Validation: Methods to ensure quality](#)

Recall task

At one-row-per-order-line grain, the pairs are (A1,1), (A1,2), (A2,1), (A2,1). Identify the duplicated composite key and explain why repeated A1 alone is not a fault. One A1 status is FALSE and the other is missing: can both be called unpaid?

DE01-04 | Data quality and useful validation

60-second recall

State the rule; inspect the value; record the result and its limit. A pass means a particular check passed. It does not prove all data is correct. Preserve source, report unknowns and investigate conflicting evidence.

[Refresh source: Data Accuracy and Validation: Methods to ensure quality](#)

Recall task

A delivery has 4 rows, 4 unique IDs, valid positive quantities and category totals that reconcile. The source date is a month old, but the owner asks for today's available stock. Explain why you still cannot claim it answers the question accurately.

DE01-05 | Meet Excel: workbook, worksheet, cells and formulas

60-second recall

Workbook means file; worksheet means one grid. Check sheet and address before typing. A1:F5 is a range. A formula begins with =; a displayed result is not the formula text. Save, close and reopen to verify your working file.

[Refresh source: Chandoo - Excel Tutorial for Beginners: Learn it FAST](#)

Recall task

On Independent, find C3 and D3 without copying the Orders values. In J2 multiply the two values using a formula, then inspect the formula bar. Keep this result separate from Orders!J2.

DE01-06 | Enter, save and structure tabular data safely

60-second recall

Choose intended types before entry or import. Keep IDs as text, numeric measures as numbers, unambiguous dates and real Boolean answers. Leave missing cells missing. Formatting changes presentation. Preserve Raw and keep the full XLSX workbook.

[Refresh source: Chandoo - Excel Tutorial for Beginners: Learn it FAST](#)

Recall task

Add order A022, Pencil, 3, 8.00, 2026-09-07, FALSE to EntryAttempt. Explain why the ID type still works and why this supplied FALSE must remain distinct from the previous blank.

DE01-07 | Sort and filter without damaging records

60-second recall

Sort complete records. Check at least one ID-to-value association after sorting. Filters hide; they do not delete. Clear filters to inspect the full input. A visible subset and a whole-range formula can cover different records.

[Refresh source: Chandoo - Excel Tutorial for Beginners: Learn it FAST](#)

Recall task

Use a fresh copy of Independent. Filter quantity to values greater than 1. List the IDs and explain whether any source record was deleted. Clear the filter and verify the full count.

DE01-08 | Basic formulas and essential functions

60-second recall

Use references for changing inputs. SUM adds; AVERAGE calculates a mean; COUNT counts numbers; COUNTA counts nonempty cells. IF chooses by a condition. SUMIF adds matching rows. Protect unknowns, inspect formulas and change one input to check that the result responds.

[Refresh source: Chandoo - Excel Tutorial for Beginners: Learn it FAST](#)

Recall task

On Retry, calculate all three order amounts. The question is now the total with a known payment status, including both paid and unpaid. Report the selected IDs, their total and unknown separately. Use your amount and payment_group columns to support the answer.

DE01-09 | Relative and absolute references

60-second recall

Relative references move when copied. Dollar signs lock the following column or row. Lock shared assumptions, let row-specific amounts move, and inspect the last copied formula. Test another input value before trusting a copied column.

[Refresh source: Excel Cell References: Relative, Absolute and Mixed](#)

Recall task

On Retry, apply a shared 12% discount to the three row amounts. State which part of the formula must move and which must remain fixed. Check the total after changing the shared rate to 0%, then restore 12%.

DE01-10 | Excel Tables and structured working habits

60-second recall

An Excel Table gives data named boundaries. Structured references name its columns. @ means this row. Inspect expansion and calculated columns. A managed rectangle can still contain wrong data, so retain key, type and reconciliation checks.

[Refresh source: Excel Cell References: Relative, Absolute and Mixed](#)

Recall task

In the Table copy, change the newly added Eraser quantity from 2 to 3. Predict the new total before looking at it. Then restore quantity 2 and save.

DE01-11 | Practical cleaning and exact lookups

60-second recall

Inspect before cleaning, state the permitted transformation and preserve the source column. Match exactly, fix the mapping range and check its keys. Unmatched or conflicting mappings are issues to explain, not invitations to guess.

[Refresh source: How to use VLOOKUP in Excel](#)

Recall task

In a copy of Lookup, deliberately add another Pen row mapped to Paper while the original Pen maps to Writing. Without accepting a first-match result as correct, describe the conflict and the evidence needed to repair it.

DE01-12 | PivotTables and Power Query awareness

60-second recall

PivotTables summarise by a chosen grain and function. Check source range, numeric types, aggregation and refresh. Power Query prepares data through recorded steps. Neither feature can repair an unclear business rule by itself.

[Refresh source: Excel Pivot Tables Tutorial for Beginners](#)

Recall task

Using the same four-row source, create a PivotTable that counts records per category instead of adding amounts. Explain why its grand total has a different unit from 400.00.

DE01-13 | Chart choice and basic visual communication

60-second recall

State the question, choose a suitable chart, label the measure and unit, check the source and respect the evidence's scope. For bars, start at zero. Exclude grand totals from category bars. Explain observed differences without inventing causes.

[Refresh source: Chandoo - Excel Tutorial for Beginners: Learn it FAST](#)

Recall task

The owner now asks for record counts: Paper 2, Writing 1, Bags 1. Create a second chart with an explicit record-count label. Explain why reusing the first chart's currency label would be wrong even though the category names match.

DE01-14 | Reconciliation, sanity checks and source preservation

60-second recall

Compare with a preserved source and state each check's purpose. Check counts, keys, values, formulas and categories. A zero difference supports one comparison. It does not erase unknowns or prove that the real-world source is accurate.

[Refresh source: Data Accuracy and Validation: Methods to ensure quality](#)

Recall task

A result keeps four rows and total 400, but swaps the recorded paid statuses of orders 0011 and 0012. Explain whether record count and whole total detect it, and calculate the new paid and unpaid amounts to show the consequence.

DE01-15 | A complete small workflow: inspect, calculate, validate, explain and save

60-second recall

Inspect the source and grain; preserve it; calculate with references; classify by explicit rules; validate independently; explain scope and unknowns; save and reopen the evidence. Passing this module shows beginner data-handling skills. It does not yet establish SQL, pipeline or job readiness. The next stage builds on this foundation.

[Refresh source: Data Journey: What you need to know for successful navigation](#)

Recall task

Use FinalRetry: D001 Pencil 4 at 9 TRUE; D002 Eraser 3 at 6 FALSE; D003 Folder 2 at 22 missing, all dated 2026-09-11. The new question asks for the amount with a known payment status, including both TRUE and FALSE, plus unknown separately. Complete a new handover and explain how the inclusion rule changed.